Fiscal Year:	FY 2025	**Task Last Updated:**	FY 02/05/2025
PI Name:	Wood, Scott J. Ph.D.		
Project Title:	Manual Crew Override of Vehicle Landings Following G-Transitions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI returned to NASA JSC in January 2017. PI was at Azusa Pacific University from August 2013 – January 2017; prior to August 2013, PI was at NASA JSC.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	04/04/2022	**End Date:**	09/30/2032
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Hannah Weiss has been added as a Co-Investigator.		
COI Name (Institution):	Duda, Kevin Ph.D. (Draper Laboratory) Wheelock, Douglas M.S. (NASA Johnson Space Center) Young, Millennia Ph.D. (NASA Johnson Space Center) Weiss, Hannah (NASA Johnson Space Center)		
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Task Description:	Manual control during exploration spaceflight consists of both planned automated supervisory control and unplanned crew override. This crew override capability is critical to enable overall mission success during landing contingencies. However, the introduction of manual override capabilities must be implemented to enable crews to mitigate risks introduced by human error. Adaptive changes in the sensorimotor system can manifest during G-transitions as spatial disorientation. While training and landing aids enable successful landing through disorientation, these adaptive changes may increase cognitive demand that need to be accounted for in the manual control strategy. It is important to characterize these effects as soon as possible following the G-transition. The primary goals of this study are (1) to understand the impact of spaceflight on crew ability to perform manual crew override tasks including the effect of flight duration, (2) to examine how adaptive changes in vestibular and cognitive function relate to changes in manual crew override proficiency, and (3) compare performance during late in-flight “just-in-time” on-board training with early post-flight crew performance. This study will recruit astronauts assigned to either short duration Private Astronaut Missions (PAM, < 30 day) or long duration (6-month) missions to the International Space Station (ISS). Ground matched-control subjects tested with the same schedule will enable us to assess the effects of learning and/or recency independent of spaceflight. Our first aim is to characterize the impact of spaceflight on crew capability to perform automated supervisory control and manual landing tasks involving crew override following both short (PAM) and longer duration ISS missions. The impact of spaceflight on piloting capability will be assessed from pre- versus post-flight changes in a simulated lunar landing during which crewmembers will manually takeover attitude and rate-of-descent to the nominal or re-designated landing aimpoint during the approach phase. This simulation will be performed on portable fixed base soon after return from ISS at the rally airport and then on R+1 using a 6DOF motion base at JSC. Subjects will also perform an automated supervisory control on a portable tablet. During this task, subjects will monitor simulation of an out-the-window display during lunar approach to determine if they need to execute a divert to an alternate landing site and use a touch screen to select the site that is within the desired landing zone and free of obstacles. The outcome measures include the percent time maintaining actual vehicle states, e.g., attitude and rate-of-descent, within recommended guidance during the landing approach, number and maximum deviation outside limits, time to execute divert to alternate landing site (if needed), error rates. The cognitive performance measures during the simulation will focus on multi-tasking in which crewmembers will respond to display prompts as a secondary task, as well as eye-head tracking to characterize visual reaction times and areas of visual attention during the simulation. We hypothesize that there will be post-flight increases in error rates and time to execute corrective actions. Our second aim is to examine how adaptive changes in vestibular and cognitive function relate to proficiency during supervisory manual control and crew override. The sensorimotor and cognitive test battery is based on a previous manual control study by Moore et al (2019) in which astronauts exhibited significant changes in motion perception, dual-tasking and sleepiness. The dual-tasking has been embedded in both tablet and piloting tasks. Motion perception accuracy and precision has been integrated with motion-base sim at JSC. Motion sickness & sleepiness will be obtained at all timepoints. We hypothesize that that a higher severity of vestibular alterations will be associated with increased flight increases in error rates and time to execute corrective actions. Given that ‘just-in-time” on-board training is an operational expectation for the extended exploration missions, all participants will perform late inflight JIT training. Crew proficiency will be captured inflight during on-board training that will be implemented on a laptop with hand controllers to allow the crewmember to practice the landing task procedures, like the approach implemented for on-board training with Shuttle landing and ISS telerobotic tasks. The metrics captured during the on-board lunar landing simulations will match those of the pre- and postflight simulations. We hypothesize that that proficiency on the “just-in-time” laptop trainer late in mission will be positively correlated with early post-flight proficiency on the same task. This project will deliver an operational demonstration of automated supervisory control and crew override capability following spaceflight and identify potential deficits that may require remediation. Comparison of individual vestibular and cognitive changes with crew override performance will help better characterize the manual control risks associated with sensorimotor alterations. Comparing performance parameters from the on-board training to post-flight performance will help demonstrate transfer of inflight on-board training to post-flight manual crew override performance. The inclusion of “just-in-time” on-board training will ensure we are characterizing changes in override proficiency with this expected countermeasure in place.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. This flight study addresses the sensorimotor research emphasis stated in the Human Research Program (HRP) Integrated Research Plan titled “Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks”. One gap associated with this risk (SM-102) is to “characterize the effects of short and long-duration weightlessness on manual control after G-transitions.” This research gap led to the solicitation of a manual control study conducted before and after long duration flights on the ISS to map changes in sensorimotor function to manual control decrements. Unfortunately, these results (Moore et al., 2019) were limited due to testing delays related to time required for direct returns from Kazakhstan, with the initial measurements conducted more than 20 hr following landing. The approach of this investigation is to leverage the commercial crew landings in the US to obtain measurements as early as possible. While the Moore study used a T-38 X-plane simulation, we will obtain measures during actual T-38 flights. We will also add a lunar landing simulation based on the current concept of operations for the HLS. Further refinements are also proposed in the sensorimotor and cognitive test battery based on the previous study.
Research Impact/Earth Benefits:	Sensorimotor function is critical for spatial orientation, gaze stabilization, and postural stability. This project examines how adaptive changes in sensorimotor and cognitive function may increase the risk of impaired ability to maintain control of vehicles and other complex systems. The goal is to map changes in physiological function with functional measures of manual control. Establishing these relationships will be relevant to how pathophysiological impairments in sensorimotor processing may affect other vehicular control tasks, such as driving with vestibular patients. This study will also be relevant to the development of strategies of how best to allow individuals with impairments perform a manual control override of autonomous driving vehicles. As Shuttle missions were extended, the on-orbit training countermeasure tool became extremely useful for commanders and pilots to practice the landing task sequence to maintain task proficiency. The ability to retain and/or assess manual control efficiency via a simple laptop-based simulation will impact our crew override approach for extended exploration missions. This approach can be easily adapted to a wide variety of simulated vehicle designs to provide similar assessments in other operational and civilian

populations.

Task Progress:

During this past year, progress was made in three key areas: (a) hardware and software integration, (b) flight study integration, and (c) supporting ground studies.
Hardware and Software Integration: Substantial progress was achieved in the software development of the Lunar Landing Simulation (LLSim) provided by The Charles Stark Draper Laboratory (Draper). The LLSim is the key component utilized across all sessions of testing in both the ground and flight studies. The LLSim is directly integrated with the 6DOF motion base, the portable simulator, and the Robotic On-Board Trainer (ROBoT) hand controllers and ROBoT computers for in-flight integration. An overview of this simulation was presented at the recent NASA Human Research Program (HRP) Investigators' Workshop (Weiss et al. 2025).

The Draper, Dynamics Skills Training (DST), and Neuroscience teams made a focused effort to develop, test, and release an integrated LLSim module within the ROBoT architecture, designed to operate with the ROBoT-specific hand controllers and 6DoF motion base across two distinct platforms: a ground simulator and inflight simulator. While both the ground and flight platforms share similar hardware, unique solutions are required to successfully provide the LLSim software across these platforms. For example, the team pursued two distinct licensing solutions for the embedded image generation software providing the lunar surface visualization. The team collaborated through two sequential ROBoT updates and nine LLSim releases to provide a functional simulator environment on a two-laptop configuration to enable the ground and flight studies. Efforts continue towards providing a DST-like workstation computer to enable testing at the postflight testing sites; example, fixed-base operator (FBO) airports.

The team is approaching certification of the LLSim for uplink to the International Space Station (ISS). A detailed certification protocol has been provided by Draper to evaluate each aspect of the LLSim software to ensure all intended actions and outcomes are supported. Preparation for testing in the Joint Station LAN (JSL) is complete, including approval for use of the ROBoT hand controllers. A target date for testing will be established when the final version of the LLSim is released from Draper. JSL testing will serve as a ground verification before uplink to Station. Once inflight, the LLSim will also be evaluated through a functional checkout. The functional checkout procedures have been generated and will be performed as non-human research by one of the crewmembers on Station. Submission of the required materials related to this non-human research have been approved by the Institutional Review Board (IRB) for implementation. The research team carefully developed a specified set of trial matrices with varying difficulty to administer across all testing sessions, encompassing preflight, inflight, and postflight sessions. Draper and DST are in the process of integrating these trial matrices into the LLSim via a patch. The integration will be completed and fully tested before the first consent crewmember begins the study. The NASA Research Operations and Integration (ROI) team facilitated communications between the research team and the NASA Marshall Space Flight Center (Marshall) Payload Operations Integration Center (POIC) to evaluate the LLSim display and user interface for adherence to the human factors standards and NASA Johnson Space Center (JSC) operations nomenclature database. Final approval from the POIC team is pending.

We collaborated with the Simulation and Graphics Branch of the NASA Engineering Directorate to develop an iOS-based multi-attribute supervisory control lunar tablet task (McDonnell et al. 2025). The Simulations and Graphics Branch (ER7) provided videos of thirty lunar landing scenarios modulating various landings sites, sun angles, camera modes, landing hazards, and navigation bias. An engineering Pathways intern integrated these videos into an iOS-based application, wrote algorithms for scenario selection, generated dual-tasking user interface elements, and developed data offload procedures for dependent measures of interest. The application is currently on the eighteenth iteration and is in a functional state to support testing. The code base has been transferred to the NASA Johnson Space Center Neuroscience Laboratory for maintenance. Future planned updates to the code base and application include a post-task summary of user performance and modifications to the user interface to promote easier scenario and trial selections. Note that the iOS platform was chosen to be compatible with tablets available in the SpaceX Dragon to leverage earlier postflight landing data collections on either ISS or free-flyer missions. A similar supervisory task will be developed for the Orion capsule during early NASA Artemis missions as part of a separate study.

We are finalizing a vestibular nulling task on the 6DoF motion base with visual vection to incorporate into the sensorimotor test battery administered during the preflight and postflight testing sessions of the flight study. The development of this capability is an extension of the initial work provided by a Master’s Degree student within the Neuroscience Lab (Bollinger 2024). This will include nulling motion with visual vection to better duplicate conditions on Apollo, where blowing dust obscured visibility during descent and appeared to contribute to variability in nulling horizontal rates, similar to brown-out disorientation during helicopter hover (Wagner 2006).

Flight Study Integration: During this past year, the Select for Flight change request (CR) was approved by the HRP Control Board. ROI facilitated the creation of the experiment document (ED) for review and approval by the Configuration Control Board (CCB). Approval was obtained in January 2025. Informed crew briefings (ICB) were provided to Crew 11 and Starliner-1. The first consent was obtained from a Crew-11 crewmember. Preflight testing sessions for the first crewmember and ICBs for Crew 12 are scheduled in February 2025. The research and ROI team completed a test readiness review (TRR) for the flight study. The final TRR approval will be obtained before the first crew test.

Supporting Ground Studies: We designed a well-defined testing protocol for the ground study to evaluate the simulated lunar landing performance of thirty healthy non-astronaut volunteers across five distinct testing sessions to inform learning progression and establish a baseline of performance. A set of LLSim trial matrices were designed to enhance participant learning and task proficiency, while maintaining a balance in task difficulty. These trial matrices were provided to Draper and DST for integration into the LLSim in the form of a patch. The results from this study seek to inform and improve efficiency of the familiarization and training sessions for the flight study.

The team managed IRB modifications and obtained approval for the ground study in October 2024. The research team also performed a test readiness review (TRR) with the NASA JSC Building 21 TRR Board, test safety officer (TSO), and medical monitor. TRR approval for the ground study was obtained in January 2025 and human testing began shortly after. Forty testing sessions are targeted to be completed before the first session of the first consented crewmember. Specialized code development is ongoing to analyze the various datasets. The ground study has been an excellent testing platform for both the hardware and software, helping identify any updates that need to be made before crew testing.

Bibliography Type:	Description: (Last Updated: 02/20/2026)
Abstracts for Journals and Proceedings	Weiss HM, Bollinger AM, Duda KR, Wheelock DH, Wood SJ. "Manual crew override of vehicle landings following g-transitions." 2025 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	McDonnell MD, Weiss HM, Gentile J, Tooher K, Hunt T, Herring M, Weno B, Wood SJ. "Supervisory control with dual tasking in post g-transition vehicle landings." 2025 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Dissertations and Theses	Bollinger AM. "Optimizing noisy galvanic vestibular stimulation for enhancing vestibular perception and manual control performance." Dissertation, University of Houston, May 2024. , May-2024